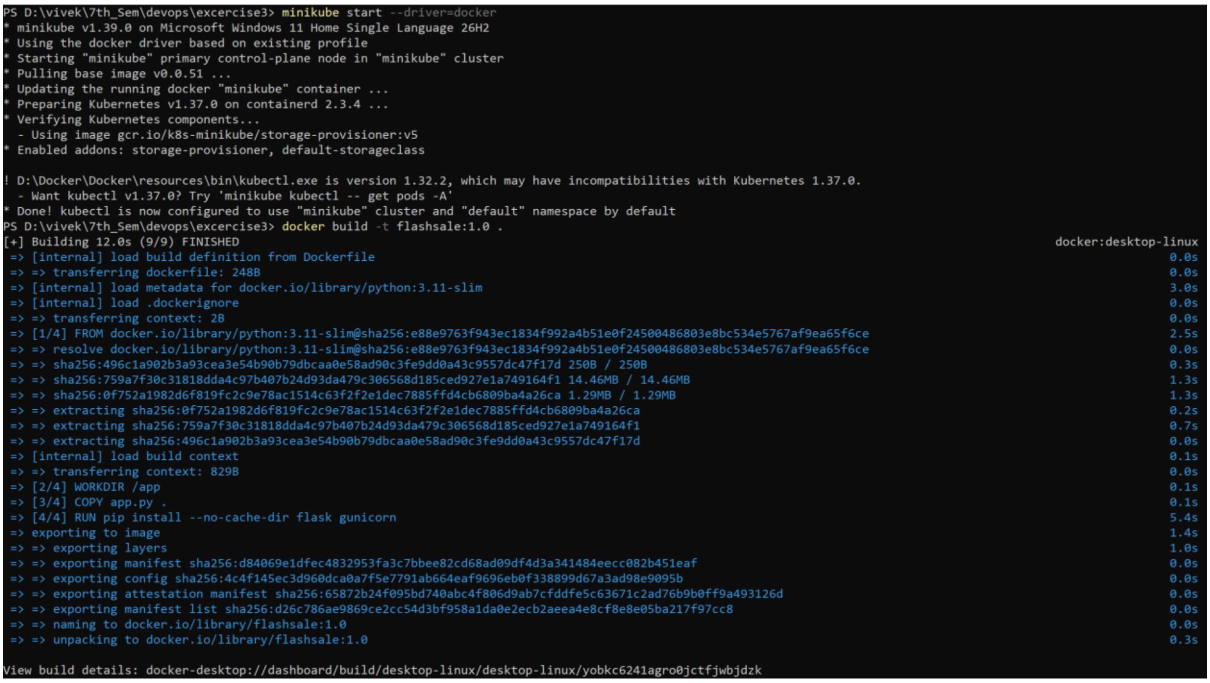
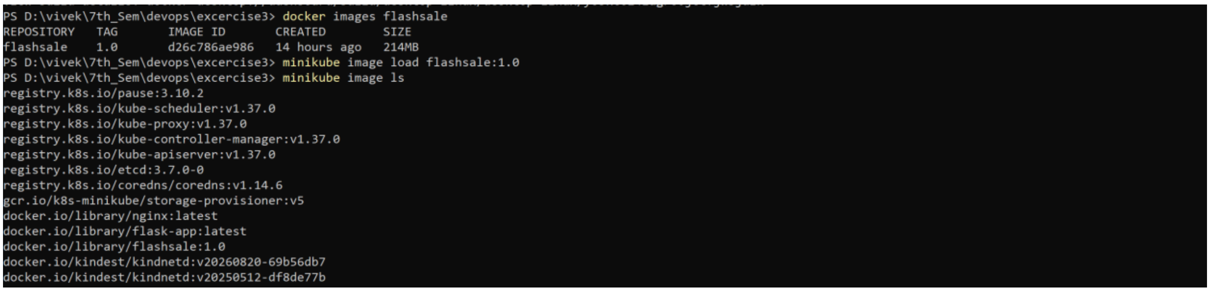
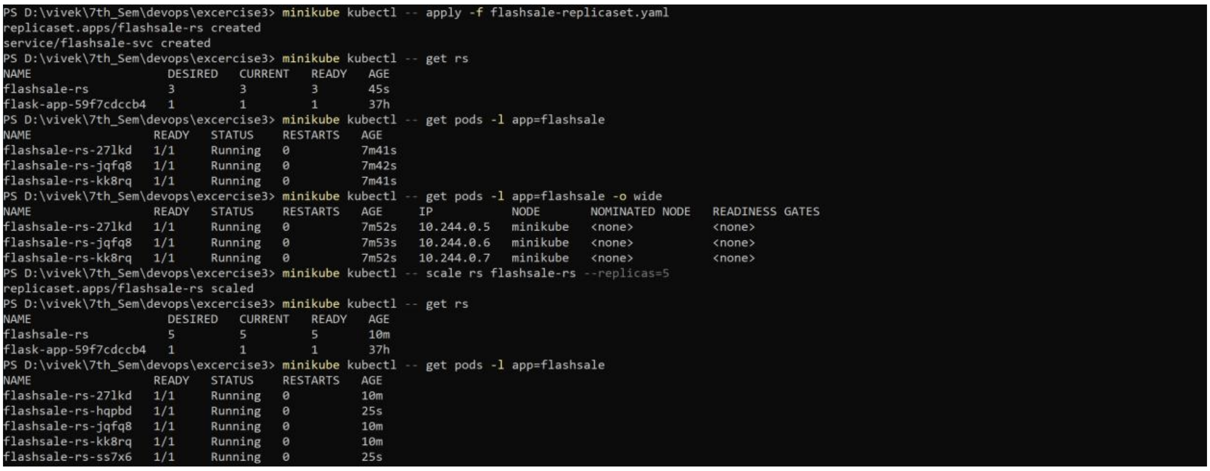
Devops – Exercise 3



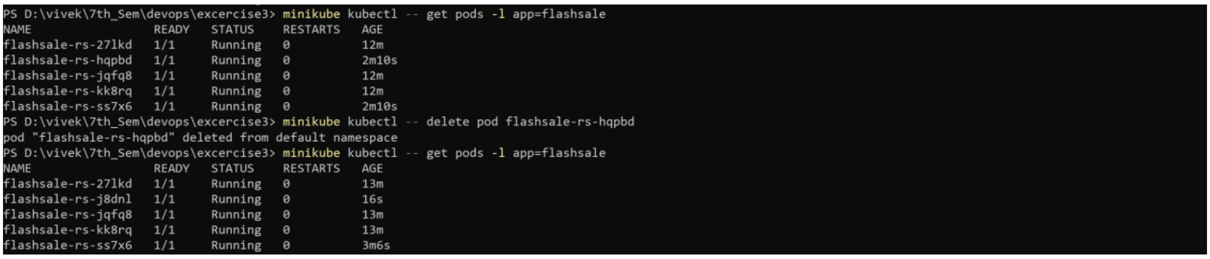
```
PS D:\vivek\7th_Sem\devops\excercise3> minikube start --driver=docker
* minikube v1.39.0 on Microsoft Windows 11 Home Single Language 26H2
* Using the docker driver based on existing profile
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.51 ...
* Updating the running docker "minikube" container ...
* Preparing Kubernetes v1.37.0 on containerd 2.3.4 ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: storage-provisioner, default-storageclass

! D:\Docker\Docker\resources\bin\kubectl.exe is version 1.32.2, which may have incompatibilities with Kubernetes 1.37.0.
  - Want kubectl v1.37.0? Try 'minikube kubectl -- get pods -A'
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
PS D:\vivek\7th_Sem\devops\excercise3> docker build -t flashsale:1.0 .
[+] Building 12.0s (9/9) FINISHED                                                                                                             docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                                                          0.0s
 => => transferring dockerfile: 248B                                                                                                                          0.0s
 => [internal] load metadata for docker.io/library/python:3.11-slim                                                                                           3.0s
 => [internal] load .dockerignore                                                                                                                             0.0s
 => => transferring context: 2B                                                                                                                               0.0s
 => [1/4] FROM docker.io/library/python:3.11-slim@sha256:e88e9763f943ec1834f992a4b51e0f24500486803e8bc534e5767af9ea65f6ce                                     2.5s
 => => resolve docker.io/library/python:3.11-slim@sha256:e88e9763f943ec1834f992a4b51e0f24500486803e8bc534e5767af9ea65f6ce                                     0.0s
 => => sha256:496c1a902b3a93cea3e54b90b79dbcaa0e58ad90c3fe9dd0a43c9557dc47f17d 250B / 250B                                                                    0.3s
 => => sha256:759a7f30c31818dda4c97b407b24d93da479c306568d185ced927e1a749164f1 14.46MB / 14.46MB                                                              1.3s
 => => sha256:0f752a1982d6f819fc2c9e78ac1514c63f2f2e1dec7885ffd4cb6809ba4a26ca 1.29MB / 1.29MB                                                                1.3s
 => => extracting sha256:0f752a1982d6f819fc2c9e78ac1514c63f2f2e1dec7885ffd4cb6809ba4a26ca                                                                     0.2s
 => => extracting sha256:759a7f30c31818dda4c97b407b24d93da479c306568d185ced927e1a749164f1                                                                     0.7s
 => => extracting sha256:496c1a902b3a93cea3e54b90b79dbcaa0e58ad90c3fe9dd0a43c9557dc47f17d                                                                     0.0s
 => [internal] load build context                                                                                                                             0.1s
 => => transferring context: 829B                                                                                                                             0.0s
 => [2/4] WORKDIR /app                                                                                                                                        0.1s
 => [3/4] COPY app.py .                                                                                                                                       0.1s
 => [4/4] RUN pip install --no-cache-dir flask gunicorn                                                                                                       5.4s
 => exporting to image                                                                                                                                        1.4s
 => => exporting layers                                                                                                                                       1.0s
 => => exporting manifest sha256:d84069e1dfec4832953fa3c7bbee82cd68ad09df4d3a341484eecc082b451eaf                                                             0.0s
 => => exporting config sha256:4c4f145ec3d960dca0a7f5e7791ab664eaf9696eb0f338899d67a3ad98e9095b                                                               0.0s
 => => exporting attestation manifest sha256:65872b24f095bd740abc4f806d9ab7cfddfe5c63671c2ad76b9b0ff9a493126d                                                 0.0s
 => => exporting manifest list sha256:d26c786ae9869ce2cc54d3bf958a1da0e2ecb2aeea4e8cf8e8e05ba217f97cc8                                                        0.0s
 => => naming to docker.io/library/flashsale:1.0                                                                                                              0.0s
 => => unpacking to docker.io/library/flashsale:1.0                                                                                                           0.3s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/yobkc6241agro0jctfjwbjdzk
```



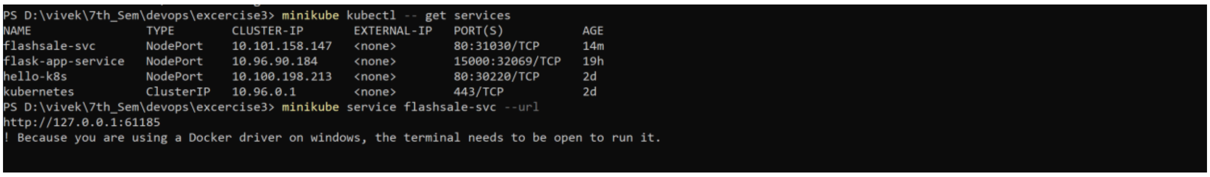
```
PS D:\vivek\7th_Sem\devops\excercise3> docker images flashsale
REPOSITORY   TAG       IMAGE ID       CREATED        SIZE
flashsale    1.0       d26c786ae986   14 hours ago   214MB
PS D:\vivek\7th_Sem\devops\excercise3> minikube image load flashsale:1.0
PS D:\vivek\7th_Sem\devops\excercise3> minikube image ls
registry.k8s.io/pause:3.10.2
registry.k8s.io/kube-scheduler:v1.37.0
registry.k8s.io/kube-proxy:v1.37.0
registry.k8s.io/kube-controller-manager:v1.37.0
registry.k8s.io/kube-apiserver:v1.37.0
registry.k8s.io/etcd:3.7.0-0
registry.k8s.io/coredns/coredns:v1.14.6
gcr.io/k8s-minikube/storage-provisioner:v5
docker.io/library/nginx:latest
docker.io/library/flask-app:latest
docker.io/library/flashsale:1.0
docker.io/kindest/kindnetd:v20260820-69b56db7
docker.io/kindest/kindnetd:v20250512-df8de77b
```



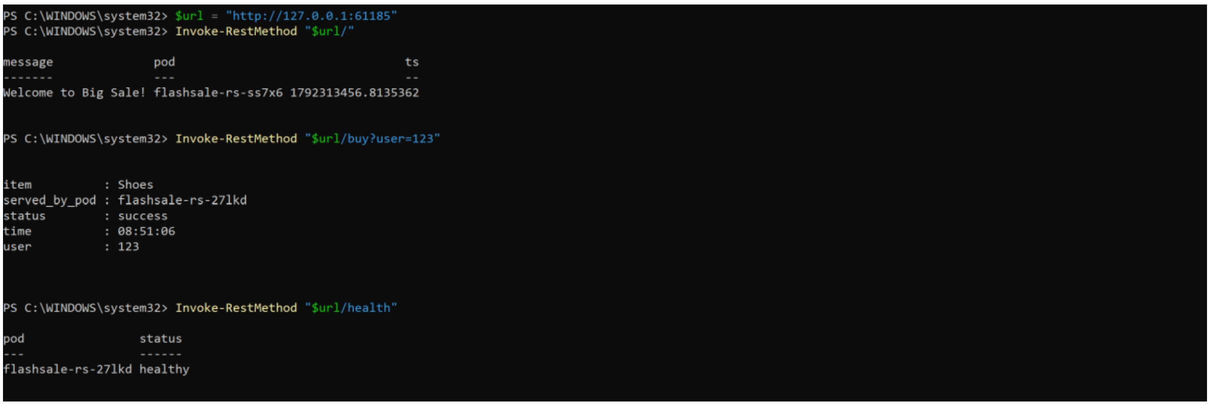
```
PS D:\vivek\7th_Sem\devops\excercise3> minikube kubectl -- apply -f flashsale-replicaset.yaml
replicaset.apps/flashsale-rs created
service/flashsale-svc created
PS D:\vivek\7th_Sem\devops\excercise3> minikube kubectl -- get rs
NAME                   DESIRED   CURRENT   READY   AGE
flashsale-rs           3         3         3       45s
flask-app-59f7cdccb4   1         1         1       37h
PS D:\vivek\7th_Sem\devops\excercise3> minikube kubectl -- get pods -l app=flashsale
NAME                 READY   STATUS    RESTARTS   AGE
flashsale-rs-27lkd   1/1     Running   0          7m41s
flashsale-rs-jqfq8   1/1     Running   0          7m42s
flashsale-rs-kk8rq   1/1     Running   0          7m41s
PS D:\vivek\7th_Sem\devops\excercise3> minikube kubectl -- get pods -l app=flashsale -o wide
NAME                 READY   STATUS    RESTARTS   AGE     IP           NODE       NOMINATED NODE   READINESS GATES
flashsale-rs-27lkd   1/1     Running   0          7m52s   10.244.0.5   minikube   <none>           <none>
flashsale-rs-jqfq8   1/1     Running   0          7m53s   10.244.0.6   minikube   <none>           <none>
flashsale-rs-kk8rq   1/1     Running   0          7m52s   10.244.0.7   minikube   <none>           <none>
PS D:\vivek\7th_Sem\devops\excercise3> minikube kubectl -- scale rs flashsale-rs --replicas=5
replicaset.apps/flashsale-rs scaled
PS D:\vivek\7th_Sem\devops\excercise3> minikube kubectl -- get rs
NAME                   DESIRED   CURRENT   READY   AGE
flashsale-rs           5         5         5       10m
flask-app-59f7cdccb4   1         1         1       37h
PS D:\vivek\7th_Sem\devops\excercise3> minikube kubectl -- get pods -l app=flashsale
NAME                 READY   STATUS    RESTARTS   AGE
flashsale-rs-27lkd   1/1     Running   0          10m
flashsale-rs-hqpbd   1/1     Running   0          25s
flashsale-rs-jqfq8   1/1     Running   0          10m
flashsale-rs-kk8rq   1/1     Running   0          10m
flashsale-rs-ss7x6   1/1     Running   0          25s
```



```
PS D:\vivek\7th_Sem\devops\excercise3> minikube kubectl -- get pods -l app=flashsale
NAME                 READY   STATUS    RESTARTS   AGE
flashsale-rs-27lkd   1/1     Running   0          12m
flashsale-rs-hqpbd   1/1     Running   0          2m10s
flashsale-rs-jqfq8   1/1     Running   0          12m
flashsale-rs-kk8rq   1/1     Running   0          12m
flashsale-rs-ss7x6   1/1     Running   0          2m10s
PS D:\vivek\7th_Sem\devops\excercise3> minikube kubectl -- delete pod flashsale-rs-hqpbd
pod "flashsale-rs-hqpbd" deleted from default namespace
PS D:\vivek\7th_Sem\devops\excercise3> minikube kubectl -- get pods -l app=flashsale
NAME                 READY   STATUS    RESTARTS   AGE
flashsale-rs-27lkd   1/1     Running   0          13m
flashsale-rs-j8dnl   1/1     Running   0          16s
flashsale-rs-jqfq8   1/1     Running   0          13m
flashsale-rs-kk8rq   1/1     Running   0          13m
flashsale-rs-ss7x6   1/1     Running   0          3m6s
```



```
PS D:\vivek\7th_Sem\devops\excercise3> minikube kubectl -- get services
NAME                TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)           AGE
flashsale-svc       NodePort    10.101.158.147   <none>        80:31030/TCP      14m
flask-app-service   NodePort    10.96.90.184     <none>        15000:32069/TCP   19h
hello-k8s           NodePort    10.100.198.213   <none>        80:30220/TCP      2d
kubernetes          ClusterIP   10.96.0.1        <none>        443/TCP           2d
PS D:\vivek\7th_Sem\devops\excercise3> minikube service flashsale-svc --url
http://127.0.0.1:61185
! Because you are using a Docker driver on windows, the terminal needs to be open to run it.
```



```
PS C:\WINDOWS\system32> $url = "http://127.0.0.1:61185"
PS C:\WINDOWS\system32> Invoke-RestMethod "$url/"

message              pod                                ts
-------              ---                                --
Welcome to Big Sale! flashsale-rs-ss7x6 1792313456.8135362


PS C:\WINDOWS\system32> Invoke-RestMethod "$url/buy?user=123"


item          : Shoes
served_by_pod : flashsale-rs-27lkd
status        : success
time          : 08:51:06
user          : 123



PS C:\WINDOWS\system32> Invoke-RestMethod "$url/health"

pod                status
---                ------
flashsale-rs-27lkd healthy
```